

Full length article

Radio frequency fingerprinting identification for Zigbee via lightweight CNN

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ABSTRACT

Zigbee is a popular communication protocol in the Internet of things (IoT) which shows great potential in smart home. However, the smart device has the risk of being hijacked by unauthorized users and may result in privacy disclosure. Traditional device identification is based on cryptography which is easy to be cracked. Recently, radio frequency fingerprinting identification (RFFID) is popular in device identification. Traditional RFFID's power consumption and cost is unacceptable to Zigbee. In order to reduce the cost, more effective model can be used to reduce the number of neurons. This paper proposes a RFFID method based on lightweight convolution neural network (CNN) which can adopt low power consumption and cost. The simulation result shows that this method can identification Zigbee device, and the accuracy reached 100%. Also, the parameter has reduced to about 93%.

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1. Introduction

With the development of communication technique, IoT makes our life become smarter and more convenient. However, smart device also causes privacy disclosure. Traditional IoT identification is based on cryptography. It uses complex algorithms to encrypt the Identification code. This method is easy to be cracked by hostile people [1,2]. Once the hacker succeeds, they can add a variety of malicious software in the device to achieve data interception and theft. It may cause terrible result, and limit the implementation of Zigbee. This crisis has led to the development of identification technology using radio frequency fingerprinting (RFF) in the physical layer. RFF is a feature that can represent a device steadily and separately. Thus, it can be thought as the DNA of device. RFF appears as carrier frequency offset (CFO), In-phase (I) and Quadrature (Q) offset, non-linear variation. It is based on the minor difference of transmitting device [3]. Even the same batch of products from the same manufacturer will be different [4]. It is hard to collect the RFF features, although the RFF has great benefits.

Tradition radio frequency fingerprinting identification (RFFID) methods are usually based on support vector machine (SVM) or decision tree (DT), random forest (RF) which are the classical machine learning methods. Restricted by the performance of the machine learning methods mentioned above, the dataset for training needs complex pre-processing [5–7]. In order to reduce

the complexity of data processing, neural network is more and more used in this field. Deep learning has showed wonderful performance in identification area. CNN is a common model that usually be used in identification [8]. However, most of method above are unavailable to Zigbee for the cost and power consumption. This paper proposed a RFFID method based on lightweight CNN. The lightweight CNN model can be implemented in Zigbee devices.

In 2003, Hall et al. from Canada proposed for the first time in literature [9] that individual identification of Bluetooth devices could be achieved by extracting subtle differences in Bluetooth signals, and defined such subtle features as “RF fingerprint”. In 2005, Ureten, O. et al. in literature [10] divided RFID into three stages: transient detection, feature extraction and fingerprint classification. In 2006, Barbuau, M. et al. introduced in the literature [11] an intrusion detection system against man-in-the-middle attacks using RF fingerprint identification technology. In 2007, Ureten, O. et al. in literature [12] divided RFID into four steps: pretreatment, detection, feature extraction and classification, and introduced a complete Wi-Fi RF fingerprint identification system. In 2008, Suski, W.C. et al. divided RFID fingerprint recognition into four steps in literature [13]: waveform parameter extraction, transient detection, feature extraction and classification of unknown received signals, and proposed to carry out individual identification of three IEEE802.11a communication devices based on OFDM signal spectrum characteristics. In 2009, Danev, B. et al. identified the node through the transient characteristics of RF signals emitted by the wireless sensor node in literature [14]. In 2009, Klein, R. W. et al. proposed in literature [15] that RF fingerprint identification is one of the physical

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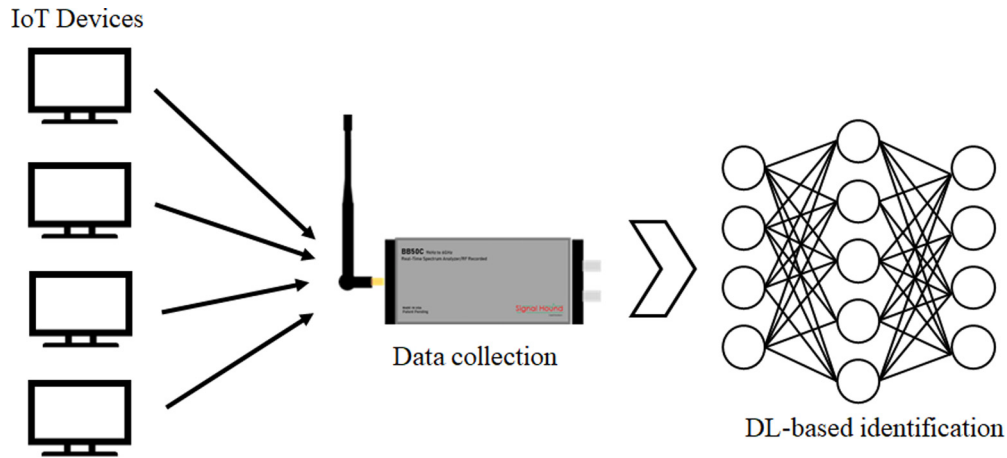


Fig. 1. System model of proposed method.

layer security technologies, which uses the unique and difficult to clone RF characteristics existing in specific communication equipment. In this paper, the recognition of 802.11a communication equipment is studied based on wavelet transform. In 2010, Danev, B. et al. proposed in literature [16] that the target of RFID is the feature recognition communication equipment based on the physical layer of communication equipment. In 2016, some colleagues proposed that on the wireless communication equipment of radio frequency finger print extraction and recognition method, has carried on the comprehensive analysis on the characteristics of fingerprint classification, summarizes the RF fingerprint should have five characteristics, universality, uniqueness, short time invariance, independence and robustness, and points out that the RF fingerprint identification is a kind of through the analysis of wireless communication signals to extract equipment of RF fingerprint recognition method, it is a method to protect the security of the communication system of the physical layer, different communication equipment of radio frequency fingerprint, can be used for communication equipment identification and access authentication.

Aiming at the identification problem of Universal Mobile Telecommunications System (UMTS) User devices, Kennedy et al. innovatively transformed the steady-state signal, namely the leading signal, of the universal mobile communication system into the spectrum as the FRF fingerprint in the literature [17] published in 2008. The experimental results show that the correct recognition rate of seven UMTS user devices with different models is as high as 91% under 15 dB SNR in laboratory environment. When 20 UMTS user devices to be identified belong to 10 of the same models, the correct recognition rate is 85%. For recognition, IEEE 802.11b equipment Brik et al. in literature [18] published in 2008, the use of more than 130 wireless card for experiment, and points out that the wireless network card equipment in the difference exists in the process of producing process, can extract signal wireless network frame, and map it to the modulation domain, the related features of these differences can be shown, and then design a passive radio frequency identification system equipment, and use of machine learning algorithms to realize the function, the experimental results show that the correct recognition rate of IEEE 802.11b devices used in experiments can reach 99%. In 2014, Peng et al. argued in literature [19] that in the field of relay authentication and recognition of collaborative relay systems, the I/Q imbalance of modulation domain could be used as a RF fingerprint of a device. This idea is verified by numerical simulation based on the authentication of two-parameter hypothesis test and likelihood ratio test. Finally, the simulation results prove the correctness of the idea, and the

authentication performance of the system is improved obviously. In 2015, aiming at the classification problem of IEEE 802.15.4 2.4 GHz RF equipment in Si Labs, Knox et al. collected and demodulated signals from five devices of this type from the same manufacturer in literature [20], and took Ettus Labs USRP1 software radio platform as the receiving device with a sampling rate of 4M. The phase information of baseband signal is extracted after filtering and noise reduction and used as RF fingerprint. The experimental results show that the classification performance of the RF fingerprint varies with temperature difference and channel distance difference. The shorter the channel distance, the higher the average classification accuracy. For the identification of RFID devices and e-passports, Danev et al. proposed a variety of fingerprint extraction methods in literature [21] and further classification and identification by using the nearest neighbor algorithm and SVM method. Experimental results showed that the final false positives rate was less than 5%, proving the effectiveness of this method. According to the existing literature, since the hardware information of the communication equipment transmitter is more stored in the steady state signal, the fingerprint characteristics of the steady state signal can be analyzed to achieve better recognition performance.

The rest of the paper are organized as follow. Section 2 introduces the structure of system model and dataset; Section 3 introduces the structure of neural network; Section 4 introduces the simulation result of our proposed method. Section 5 makes a conclusion of our work.

2. System model and data collect

2.1. System model

The system model of the proposed method is show as Fig. 1. As can be seen in the figure, the signal of Zigbee is collected to build a dataset of RFF. A CNN is used to identification the sampling signal, and then the model was pruned to get a smaller model.

2.2. Data collect

Then the signal of devices was collected by Spectrum analyzer (BB50C) from Signal Hound. The dataset in this paper comes from 4 Zigbee devices. Each device has collected about 60,000 samples. In addition, each sample has 1000 sampling points.

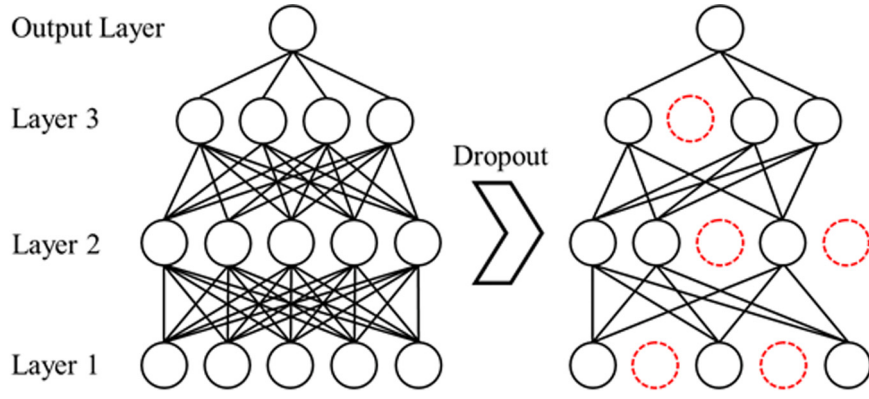


Fig. 2. Schematic diagram of dropout.

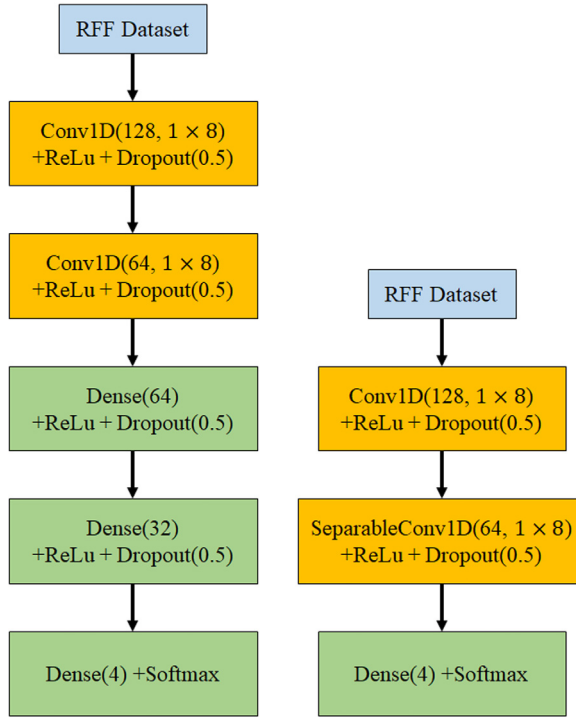


Fig. 3. The structure of neural network.

3. The proposed lightweight method

3.1. Structure of lightweight CNN

CNN is a classical model in the area of image identification, it also shows wonderful performance in feature extraction. CNN usually consist of convolution layers, pooling layers, dense layers. The detail descriptions of those layers are as follow:

Convolution layer: This layer is used to extract the feature of the dataset which can replace the role of data pre-process in most cases. By using the convolution kernel, the neural network reduces the dimension of input data with the valuable feature collected. Thus, with the help of convolution layer, many complex data-processing can be avoided.

Pooling layer: Pooling layer also named down-sampling layer. It usually be used to reduce the number of parameters. Max pooling and average pooling are common methods of pooling. However, pooling layers will also reduce the accuracy of neural networks.

Dense layer: Dense layer is used to map the feature space to sample space, which is necessary in classification model. However, before the features be send into the Dense layer, they must be flattened. This results in the increase of parameters and training times.

The structure of CNN and lightweight CNN used in this paper is as Fig. 2. In this paper, the neural network adopts convolution layers and dense layers. In consideration of the size of input dataset, there is no need to use the pooling layers. The ReLU and Softmax is the popular nonlinear activation function in deep learning. The purpose of deep learning is to use a pile of neurons to pile up the general appearance of a function, and then through a large number of data to reverse fit the various parameters of this function, and finally outline the complete shape of the function. Thus, if a neural network only has linear activation function, the expressive ability of it will be limited. The formulas of them [22–27] are given as follows:

$$f_{ReLU}(z_i) = \max(z_i, 0), \quad f_{Softmax}(z_i) = \frac{e^{z_i}}{\sum e^{z_i}} \quad (1)$$

where $\max(\cdot)$ means choose the bigger in the function. The z_i is the training sample. ReLu usually be used between layers, while Softmax is a classical function in multi-classification problem. Comparing these two functions, it can be seen that the Softmax has a “Gradient Disappear” problem. Obviously, the gradient of the Softmax in $|z| > 4$ is very small. In addition, the calculation process of ReLU is much faster than that of Softmax.

The performance of ReLU in sparsity problem is very great, it makes each neuron can maximize its role in screening, in line with a certain feature of the intermediate value will be enlarged a lot; inconsistent will be directly deleted. However, when the gradient is too large and the learning rate is set too large, the weight will be updated too much at once. To any dataset the z_i will be a negative number. Then the output of neural network with ReLU will be zero. In this case, the ReLU can be represented by Leaky ReLu. From the formulas, the sum of the values output by Softmax is 1, This makes Softmax suitable for multi-classification. For example, if the neural network is used to identify target data with 4 categories. The Softmax can output a vector with 1×4 size. The vector can be seen as the possibility of which category the data belong to. The Dropout is used to make random neurons “dead” in order to reduce the possibility of over-fitting. The parameter of it represent the rate of “dead” neurons in “alive” neurons [28–31]. The process is shown in Fig. 4. Dropout randomly “killed” some neurons which reduce the possibility of neural network over dependence on some local features to fit the function.

As show in left of Fig. 3, the common CNN adopts two convolution layers named “Conv1D”, and three Dense layers. The sizes

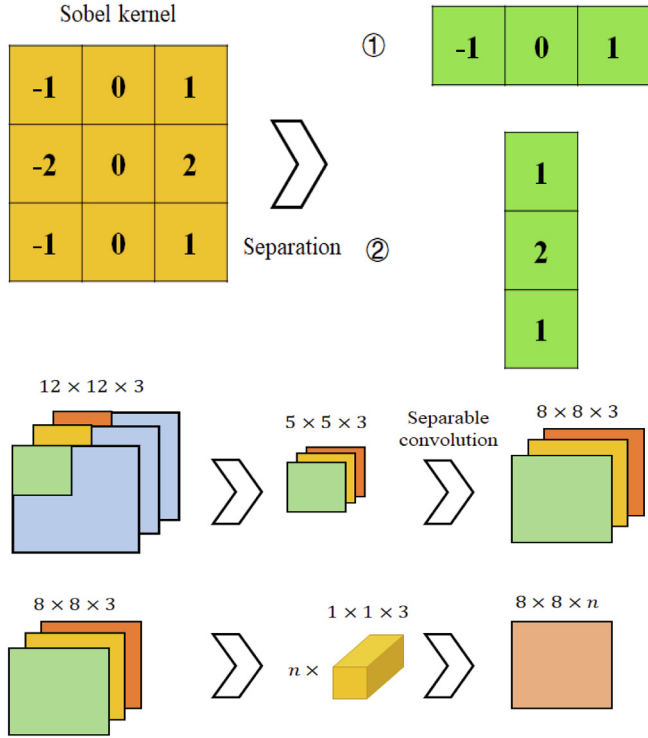


Fig. 4. The process of space separable convolution kernel.

of convolution kernels are 1×8 and 1×4 corresponding to the 1 dimension. While the right part shows the proposed lightweight CNN. Compared with the common CNN, the lightweight CNN reduces the number of Dense layers, and replace the convolution layer with “Separable CNN”. The size of kernels in the first two layer is the same. The main ideal of “Separable CNN” is split the big convolution kernel into small ones. The process is shown in Fig. 4.

“Separable CNN” usually has two methods to reduce the number of parameters called space separable convolution and depth separable convolution. As shown in Fig. 4(a), in this paper, the Sobel kernel is used to as an example to introduce the space separable convolution. Sobel kernel is a classical convolution kernel for edge detection. The Sobel kernel is an array which size is 3×3 . Through the separation, it can be split into a 1×3 vector and a 3×1 vector. Thus, when use the separable kernel, it takes 6 parameters instead of 9 parameters [32].

In Fig. 4(b), a simplified depth separable convolution is shown, the n means there are n kernels. This method split common convolution into convolution on different channels and convolution on whole channel. The core of this process is separable convolution, three kernels convolute the data in different channels separately. Then n kernels are used to convolute the data with the same number of channels to get the result. Then the result feature map is the same size of use big kernel. Actually, the example above is just a simplification. The true separation is much more difficult than these methods in [33–36].

3.2. Loss function and optimizer

The loss function and optimizer also play an important role in neural network. With suitable loss function and optimizer, the neural network will fit soon. Compared with the optimizer used before, the Adam has high computational efficiency, minimal memory requirements. In addition, It is very suitable for problems with large data or parameters, for non-fixed targets, and

Table 1

The classification performance accuracy of different methods.

Methods	Accuracy (%)
SVM	54.55
RF	77.77
DT	61.02
CNN	100
Lightweight CNN	99.99

Table 2

The computational and spatial complexity between CNN and Lightweight CNN.

Methods	CNN	Lightweight CNN
FLOPs	8,191,315	533,212 (93.49%↓)
Parameter	4,092,324	264,388 (93.54%↓)
Model size (kB)	48,001	3133 (93.47%↓)

Table 3

Dataset for training samples, validation samples and test samples.

Device No.	Training samples	Validation samples	Test samples
1	22,400	9600	30,000
2	36,400	15,600	30,000
3	26,600	11,400	30,000
4	22,400	9600	30,000

for small SNR or sparse gradient problems. The optimizer used here is Adam which is an extension of SGD, which can replace the classical stochastic gradient descent method to update the network weight more effectively. In short, Adam uses Adaptive learning rate and momentum to speed up the process of training. It can be written as

$$m_{t+1} = \beta_1 m_t + (1 - \beta_1) \nabla_{w_t} L \quad (2)$$

$$v_{t+1} = \beta_2 v_t + (1 - \beta_2) (\nabla_{w_t} L)^2, \quad (3)$$

$$\hat{m}_{t+1} = \frac{m_{t+1}}{1 - \beta_{1,t}}, \hat{v}_{t+1} = \frac{v_{t+1}}{1 - \beta_{2,t}}, \quad (4)$$

$$w_{t+1} = w_t - \eta \frac{\hat{m}_{t+1}}{\sqrt{\hat{v}_{t+1} + \epsilon}}, \quad (5)$$

The m_t means the first moment estimate; v_t means the second moment estimate; t represents training step; β_1 and β_2 are exponential decay rates for the moment estimates; $\beta_{1,t}$ and $\beta_{2,t}$ means the t th power of β_1 and β_2 . The loss function also matters to the model, big loss function will make this model more efficient. Usually, the categorical cross entropy (CCE) and the binary cross entropy (BCE) adopt most situation of identification problem. The difference of them are in the number of categories they can support. This paper chooses the CCE as the loss function. The CCE can be written as:

$$L = - \sum_{i=1}^C y_i \log f_i(x) \quad (6)$$

where x represents the input data; the y_i means the true label of x ; C means the number of categories; $f_i(\cdot)$ means the predictions of the model which in the identification problem means labels.

4. Simulation results

The simulation of CNN and lightweight CNN is based on Keras with Tensorflow as backend, and it is implemented with GTX1650. The dataset is pre-processed by MATLAB. The traditional machine learning methods and the confusion matrix are based on Sklearn.

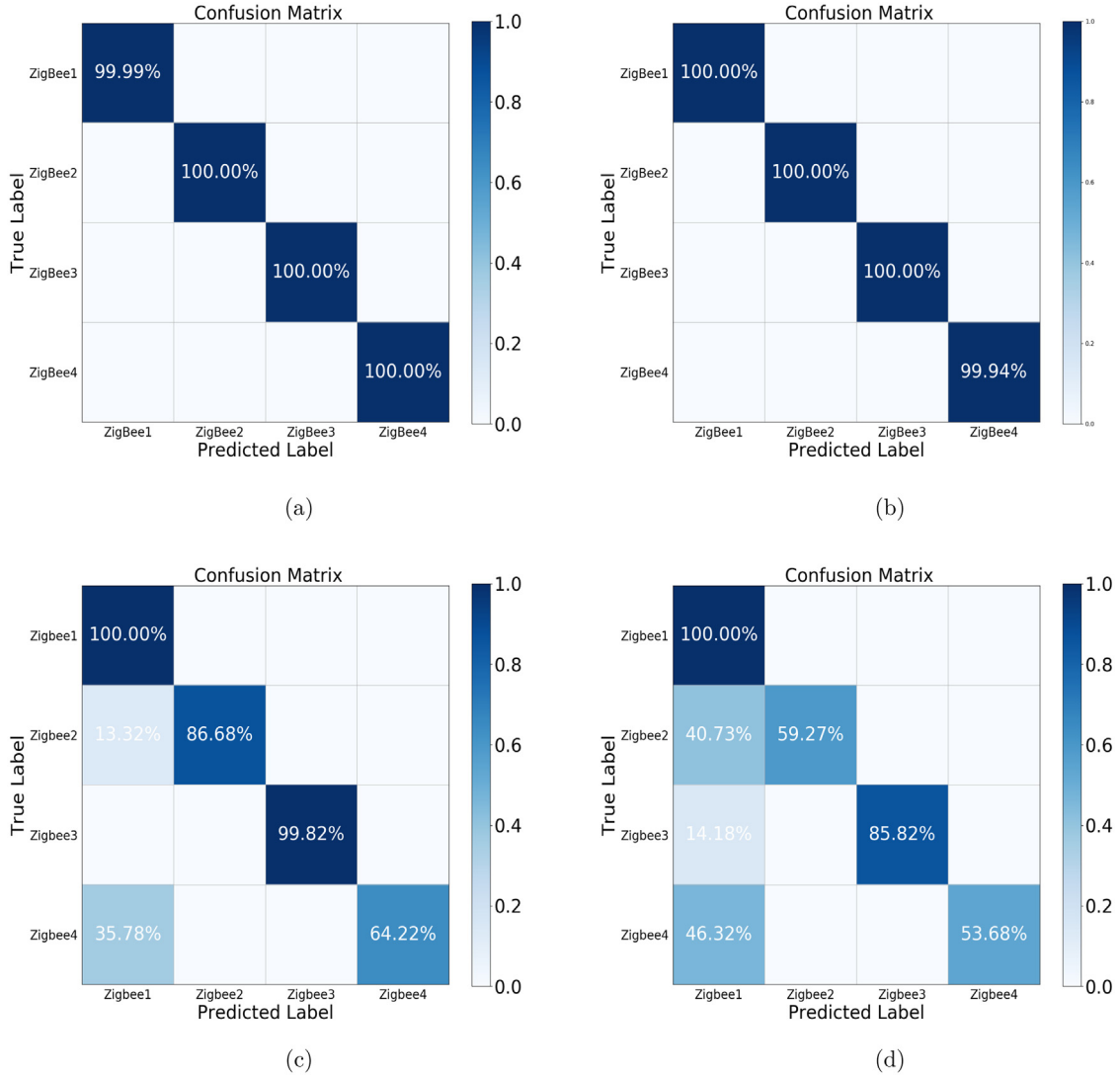


Fig. 5. The confusion matrix of CNN and lightweight CNN. (a) CNN; (b) Lightweight CNN; (c) RF; (d) DT.

4.1. Classification performance

In this article, a CNN is designed to recognize four different devices. Then, through the lightweight operation of the designed convolutional neural network, the accuracy difference of the convolutional network before and after the light-weight is compared under the condition of training the same set of datasets. In addition, this article also adopts and analyzes other three traditional machine learning algorithms, which are support vector machine (SVM), decision tree (DT), and random forest (RF). From the results, both CNN and lightweight CNN show excellent accuracy compared with the other three methods. Compared with the CNN, although the lightweight implemented in this article has a little performance loss, it still has a more concise network framework and faster running speed. The classification performance is shown in Table 1.

4.2. Complexity analysis

As mentioned above, although lightweight CNN has slightly worse performance than CNN, its network complexity is much lower than CNN. Detailed data is shown in Table 2. As is shown in the table, this article provides three metrics: floating-point operations (FLOPs), parameters and model size. The former metrics are

used to indicate the complexity of the calculation, while the latter two metrics are used to show the complexity of the network structure.

The other two indicators can be easily understood, while the meaning of FLOPs cannot be understood at a glance. FLOPs means the number of floating-point operations which is the main calculation in neural network, which is understood as the number of calculation, and it has nothing to do with hardware. Thus, the FLOPs usually be used to measure the complexity of model trained on different device. For example, model A use the NVIDIA GTX1060 as the implementation platform, while the model B use the NVIDIA TX2. It is hard to figure out which model is complex for the ability of the implementation platform is different. However, the number of floating-point operations will not change with the implementation platform. Thus, it can be a standard to measure the complexity of models.

The formula of FLOPs in different layers depend on the parameters of them. For a convolutional layer, suppose its size is $h \times \omega \times c \times n$ (where c represents input channel, n represents output channel), the output feature map size is $H' \times W'$, then the parameters of the convolutional layer can be calculated as follows:

$$paras = n \times (h \times \omega \times c + 1) \quad (7)$$

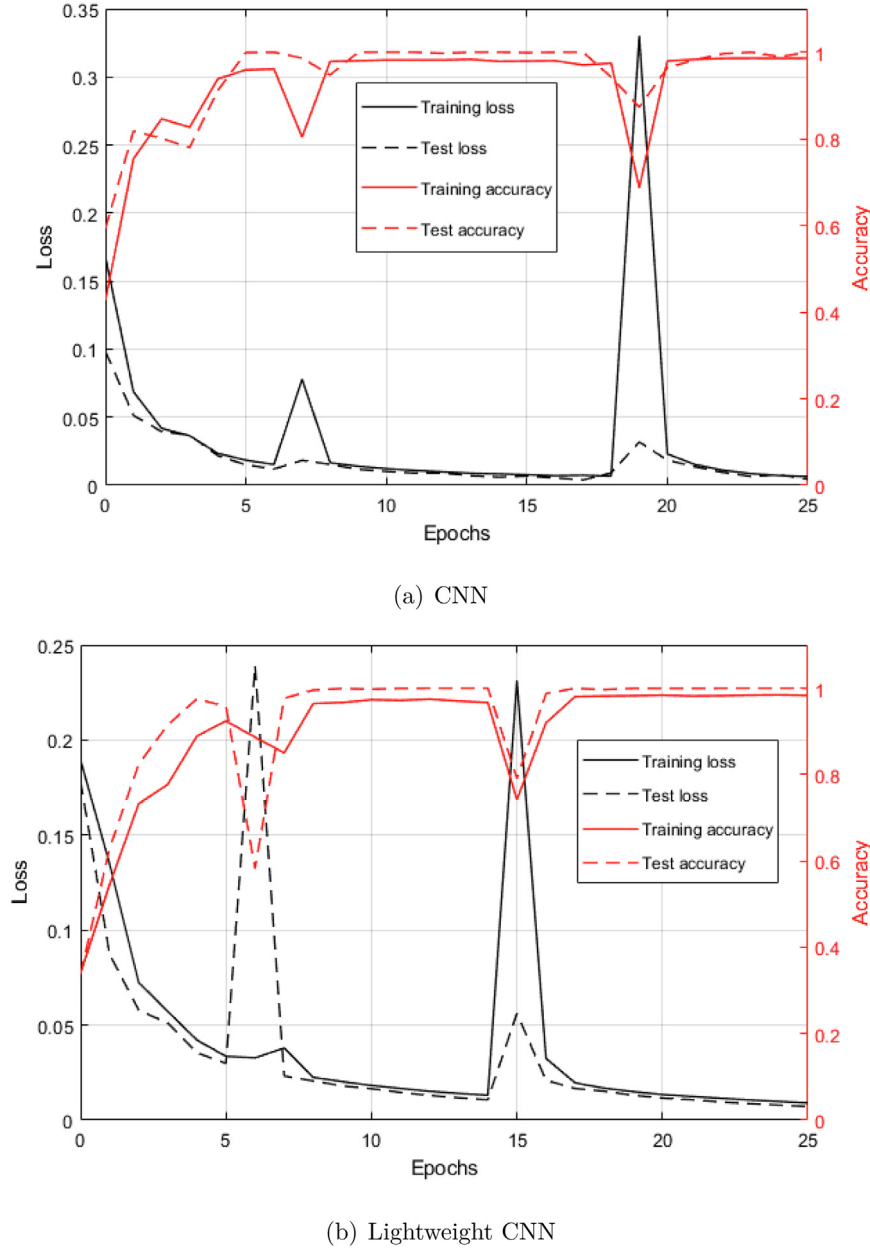


Fig. 6. Loss and accuracy vs. epoch.

$$FLOPs = H' \times W' \times n \times (h \times \omega \times c + 1) \quad (8)$$

The above two formulas can also be represented by:

$$FLOPs = H' \times W' \times paras \quad (9)$$

With the above calculation formula, the FLOPs of CNN and lightweight CNN can be calculated respectively. The specific data is shown in the Table 2. The lightweight CNN has only 6.51% computation of CNN, which shows that the computational complexity of lightweight CNN is lower than that of CNN. Which means the lightweight CNN can be used in some platform with low ability of calculation. Thus, considering the cost of training and the adaptability, the performance of lightweight CNN is much higher than that of CNN.

The Table 3 shows the size of dataset used in the paper. Each device has different number of training sample and 30,000 testing sample. 70% of training dataset are used as the train data, while

the rest of them as the validation data. As shown in Fig. 5, it is easy to see that the performance of CNN and Lightweight CNN is best, basically identifying the device entirely, with accuracy approaching 99.99%. However, the traditional machine learning method is not very effective for there is no data pre-processing. Among them, RF can identify the first device completely, and can identify almost all the third device and most of the second device. DT can recognize the first device completely and most of the third device, but it is not effective to identify the second and fourth device.

4.3. Loss and accuracy vs. epoch

The curves of loss and accuracy of the training samples and test samples are shown in Fig. 6. It is easy to get that the accuracy and the loss curves of both the training samples and test samples are very similar, both of them demonstrated the excellent performance of the network. The loss drops greatly in

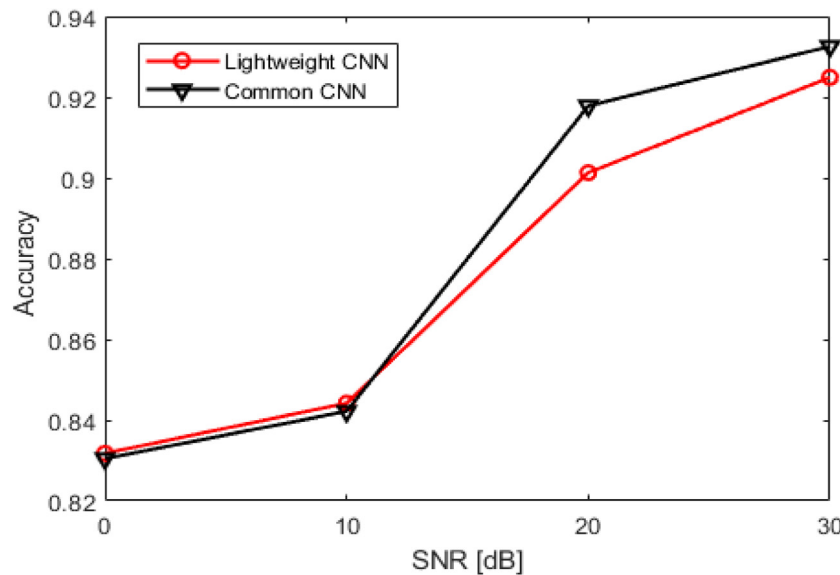


Fig. 7. Accuracy in different SNR.

several epochs, which shows that the network based on deep learning can give accurate judgments faster than those traditional methods. Though the CNN and Lightweight CNN show similar performance, the networks can be also evaluated from other sides such as operating speed. From Table 2, it is easy to draw conclusions that the Lightweight CNN shows excellent streamlined network and extremely fast running speed.

4.4. Accuracy in different SNR

As we can see in Fig. 7, the accuracy improved with the increase of SNR. In addition, the performance of common CNN is better than the lightweight CNN. Especially in the high SNR environments. The noise has a devastating effect to the system, in low SNR environments, the accuracy decreases to about 85%. For the aim of RFF identification, it is unacceptable.

5. Conclusion

This paper proposed a lightweight CNN to identify Zigbee devices and compare it with the common CNN and classical machine learning methods. Both of them has excellent performance in high SNR environment. The simulation result shows that the performance of lightweight CNN proposed is better than the machine learning methods, such as SVM, DT, the performance of it is worse than the common CNN. However, the cost of lightweight CNN proposed is much smaller than the common CNN.

CRediT authorship contribution statement

Guangwei Qing: Conceived of the presented idea, Developed the RFFID method based on lightweight convolution neural network, Discussed the results, Contributed to the final manuscript. **Huifang Wang:** Conceived of the presented idea, Conducted the simulation experiments, Verified the proposed methods, Discussed the results, Contributed to the final manuscript. **Tingping Zhang:** Conducted the simulation experiments, Verified the proposed methods, Discussed the results, Contributed to the final manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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